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PROJECT REPORT

AI Fashion Recommendation Systems and Virtual Try-On Research

Introduction

Artificial Intelligence has become an important part of the fashion industry, helping businesses provide personalized shopping experiences and improve customer satisfaction. Modern fashion platforms use recommendation systems to suggest products based on user preferences and virtual try-on technologies to allow customers to visualize outfits before making a purchase.

As part of my AI/ML Internship at Raritone, I worked on researching various AI technologies used in fashion applications. The project focused on understanding recommendation systems, virtual try-on models, datasets used for training these models, and the overall workflow involved in building intelligent fashion solutions.

This research helped me understand how Artificial Intelligence can enhance online shopping experiences by making recommendations more accurate and reducing uncertainty during purchasing decisions.

Project Overview

During my internship at Raritone, I worked on researching AI-based fashion recommendation systems and virtual try-on technologies. The project focused on understanding how Artificial Intelligence can improve online shopping experiences through personalized recommendations and digital clothing visualization.

Project Objectives

The primary objective of this project was to study the role of Artificial Intelligence in fashion recommendation and virtual try-on systems.

The specific objectives included:

- Understanding different recommendation techniques used in fashion platforms.
- Exploring virtual try-on technologies and their applications.
- Studying datasets used for training recommendation and virtual try-on models.
- Comparing different virtual try-on models and analyzing their strengths.
- Designing a workflow for a personalized fashion recommendation system.
- Understanding current trends and future opportunities in fashion AI.

Research Conducted

During the internship, extensive research was conducted on recommendation systems used by modern fashion and e-commerce platforms.

The first area of study was Content-Based Filtering. This technique recommends products based on the user's previous interactions and preferences. If a user frequently views or purchases a specific type of clothing, the system recommends similar products.

The second technique studied was Collaborative Filtering. This method recommends products based on the behavior of users with similar interests. It identifies patterns among users and provides suggestions based on shared preferences.

Hybrid Recommendation Systems were also explored. These systems combine Content-Based and Collaborative Filtering methods to improve recommendation accuracy and overcome the limitations of individual approaches.

Additionally, Deep Learning-Based Recommendation Systems were studied. These systems use neural networks to understand user behavior, fashion trends, and product relationships, enabling more personalized recommendations.

This research provided valuable insights into how AI-powered recommendation systems improve customer engagement and enhance shopping experiences.

Models Explored

A major part of the project involved studying virtual try-on technologies and understanding how different models digitally fit garments onto a person's image.

VITON was one of the first virtual try-on models explored. It introduced image-based garment transfer and established the foundation for future virtual try-on research.

CP-VTON improved garment alignment and clothing warping techniques, producing more realistic fitting results compared to earlier approaches.

HR-VITON focused on generating higher-quality outputs while preserving clothing details and textures. It demonstrated significant improvements in image quality and realism.

StableVITON was another important model studied during the research. It utilizes diffusion-based image generation techniques to create highly realistic virtual try-on results. The model showed excellent garment preservation and visual quality.

IDM-VTON was also explored as a modern virtual try-on approach. It focuses on maintaining clothing details while generating realistic outputs that closely resemble actual garment fitting.

Studying these models helped me understand the evolution of virtual try-on technology and the techniques used to improve fitting accuracy and image quality.

Datasets Studied

To understand how AI models are trained, several fashion datasets were explored during the research process.

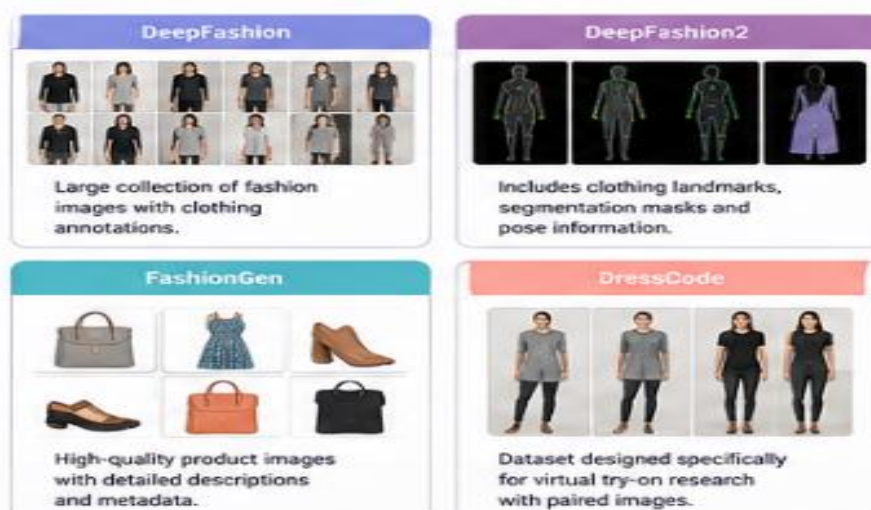
DeepFashion is one of the most widely used fashion datasets and contains a large collection of clothing images with detailed annotations. It is commonly used for clothing recognition and recommendation research.

DeepFashion2 extends the original dataset by providing additional information such as clothing landmarks, segmentation masks, and human pose annotations.

FashionGen contains high-quality fashion product images and detailed descriptions, making it useful for recommendation and image generation tasks.

DressCode is specifically designed for virtual try-on research and contains paired images of garments and models wearing them.

These datasets play an important role in training AI models and improving the performance of recommendation systems and virtual try-on applications.



2D Virtual Try-On Findings

2D virtual try-on technology focuses on generating clothing images directly on a user's photograph. These systems use computer vision and deep learning techniques to

digitally transfer garments onto a person's image while preserving the overall appearance of the clothing.

During the research, it was observed that 2D virtual try-on models are relatively simple compared to 3D approaches. They require less computational power, faster processing time, and smaller datasets, making them suitable for large-scale e-commerce platforms. Models such as VITON, CP-VTON, and HR-VITON have shown significant improvements in garment alignment, texture preservation, and image quality over time.

One of the major advantages of 2D virtual try-on systems is their efficiency. They can generate realistic outputs quickly and are easier to deploy in online shopping applications. These systems help customers visualize how a garment may look on them before making a purchase decision.

However, 2D virtual try-on also has certain limitations. Since the output is generated from a single image, body shape representation may not always be accurate. The systems may struggle with complex poses, loose garments, or situations where parts of the body are hidden. Additionally, users can only view the outfit from a limited perspective, which may reduce the realism of the experience.

Overall, 2D virtual try-on remains a practical and widely adopted solution due to its speed, simplicity, and lower implementation cost.

3D Virtual Try-On Findings

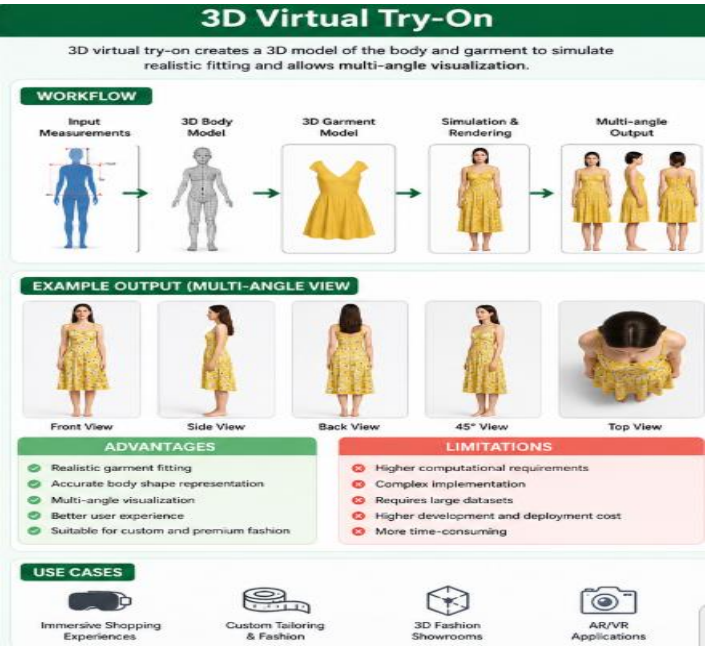
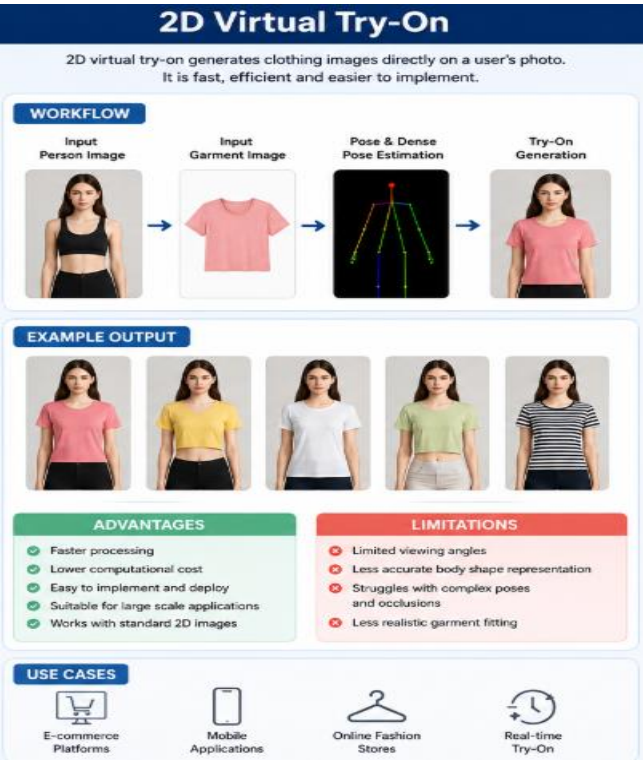
3D virtual try-on technology creates a three-dimensional representation of both the user and the garment to simulate realistic clothing fitting. These systems use body models, garment models, and physics-based simulations to generate a more immersive virtual shopping experience.

The research showed that 3D virtual try-on systems provide a much higher level of realism compared to 2D approaches. They can accurately represent body shape, garment draping, and clothing fit from multiple viewing angles. This allows users to better understand how a garment would appear and fit in real-world situations.

One of the biggest advantages of 3D virtual try-on is its ability to provide a realistic visualization experience. Users can view garments from the front, side, and back, making the shopping process more interactive and informative. These systems are particularly useful for custom clothing, luxury fashion, and advanced e-commerce applications.

Despite their advantages, 3D virtual try-on systems require significantly higher computational resources and more complex implementation techniques. Building accurate 3D body models and garment simulations requires large datasets, advanced algorithms, and powerful hardware. The development and deployment costs are also higher compared to 2D systems.

Based on the research, 3D virtual try-on represents the future direction of fashion technology due to its superior realism and user experience. As computing resources become more accessible and AI models continue to improve, 3D virtual try-on systems are expected to become more common in online fashion platforms.



Comparative Observation

Both 2D and 3D virtual try-on technologies play an important role in modern fashion applications. While 2D systems offer speed, simplicity, and lower computational requirements, 3D systems provide greater realism, accuracy, and interactivity.

For large-scale commercial platforms, 2D virtual try-on remains a cost-effective solution. However, for applications where realistic fitting and user experience are the primary goals, 3D virtual try-on offers significant advantages.

The future of fashion AI is likely to involve a combination of advanced recommendation systems and realistic virtual try-on technologies to create highly personalized and engaging shopping experiences.

Key Findings

One of the major findings from this research was that recommendation systems play a significant role in increasing customer engagement and improving personalization.

The study also revealed that virtual try-on technologies help users make more confident purchasing decisions by allowing them to visualize outfits before buying.

Among the models explored, StableVITON and IDM-VTON demonstrated highly realistic results and represented some of the most advanced approaches in virtual try-on research.

The combination of recommendation systems and virtual try-on technology has the potential to transform the future of online fashion shopping.

Skills Gained

Through this project, I gained knowledge in recommendation systems, computer vision, deep learning, virtual try-on technologies, and fashion datasets. I also improved my research, analytical thinking, and technical documentation skills.

Conclusion

The completion of this project provided me with valuable exposure to the practical applications of Artificial Intelligence in the fashion industry. Throughout the internship, I researched recommendation systems, virtual try-on technologies, fashion datasets, and AI workflows to understand how these technologies contribute to improving online shopping experiences.

The study of recommendation systems helped me understand how platforms analyze user preferences and behavior to provide personalized product suggestions. Exploring Content-Based Filtering, Collaborative Filtering, Hybrid Recommendation Systems, and Deep Learning approaches gave me insight into the techniques used to enhance customer engagement and satisfaction.

Research on virtual try-on technologies allowed me to understand how Artificial Intelligence and Computer Vision can be used to digitally fit garments onto a user's image. By studying models such as VITON, CP-VTON, HR-VITON, StableVITON, and IDM-VTON, I learned about the advancements that have been made in improving garment alignment, image quality, realism, and overall user experience.

The analysis of fashion datasets such as DeepFashion, DeepFashion2, FashionGen, and DressCode highlighted the importance of high-quality data in training and evaluating AI models. These datasets play a crucial role in improving the performance and accuracy of recommendation systems and virtual try-on applications.

One of the key takeaways from this project is that the combination of recommendation systems and virtual try-on technologies has the potential to significantly transform the future of online fashion retail. Personalized recommendations can help users discover products that match their preferences, while virtual try-on systems can reduce uncertainty by allowing customers to visualize garments before making a purchase.

In addition to technical knowledge, this internship also helped me improve my research, analytical thinking, problem-solving, and technical documentation skills. The process of studying research papers, comparing different models, and organizing findings strengthened my ability to understand and communicate complex technical concepts effectively.

Overall, this internship has been a valuable learning experience that enhanced my understanding of Artificial Intelligence, Deep Learning, and Computer Vision while providing practical insight into how these technologies are applied in real-world fashion and e-commerce solutions. The knowledge and skills gained during this project will be beneficial for my future academic and professional growth.

Acknowledgement

I am thankful to the mentors and the entire Raritone team for their continuous guidance, support, and encouragement throughout the internship. Their valuable feedback and assistance helped me successfully complete the project and enhance my technical and professional skills.